To my family, and to everyone who discussed a factor graph with me at improbable hours across improbable time zones.

Contents

Chapter 1

Introduction

1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). Their central object is the *score function* — the gradient of the log-density of progressively noised data,

$$S(x,t) \;=\; \nabla_x \log p_t(x), \tag{1.1}$$

which is all a model needs to know in order to reverse a noising process and turn pure noise back into data (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of what it has to learn remains poorly understood.

A complementary research programme, initiated in the statistical-physics community, replaces the black box with *exactly solvable models*: data distributions simple enough that the score can be computed in closed form, yet rich enough to exhibit the phenomena observed in real systems — speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024). Exact scores act as a microscope: every regime, transition time, and failure mode can be traced back to explicit formulas.

This thesis extends the exactly solvable programme in a direction that is natural for data with *temporal or sequential structure*: the data are not a cloud of independent points but a *trajectory* $a_0, a_1, \ldots, a_{K-1}$ of a Markov process — the prototype of a video, a time series, or any dynamic object. Each frame of the trajectory is corrupted independently by an Ornstein–Uhlenbeck (OU) channel, and the object of study is the *joint* score of the whole noisy sequence. Two classical bodies of theory then meet the modern one. First, the joint score of a Markov chain turns out to be governed by the algebra of structured matrices — Toeplitz covariances, tridiagonal precisions, and their spectral properties. Second, computing the score is an exercise in Bayesian inference on a tree-shaped factor graph, the home turf of *belief propagation* (BP) and its Gaussian projection, *approximate message passing* (AMP) (Pearl, 1988; Mézard and Montanari, 2009; Donoho et al., 2009). The thesis develops both routes in full detail, proves where they coincide, and measures precisely what is lost when they are forced apart.

1.2 Positioning and gap

The literature this work builds on can be organised in three strands.

Statistical physics of diffusion models. Closed-form analyses of diffusion generation exist for Gaussian mixtures, random energy models, and high-dimensional asymptotics (Biroli and Mézard, 2023; Biroli et al., 2024; Ghio et al., 2024). These works treat the data as unstructured collections of samples; the time axis they study is the *diffusion* time. The internal time of a *sequence* — the fact that a video frame is correlated with its neighbours through a dynamical rule — is absent from these analyses.

Inference on chains. Exact posterior inference on Gauss–Markov chains is classical: the Kalman filter and smoother (Kalman, 1960; Rauch et al., 1965), hidden Markov models (Rabiner, 1989), and their unifying formulation as sum–product message passing on factor graphs (Kschischang et al., 2001; Mézard and Montanari, 2009). What this literature does not ask is the question that matters for generative modelling: *how does the exact posterior — and therefore the exact score — deform as a function of the diffusion time*, and what does that deformation imply for the architecture of a network that must learn it?

Message passing with continuous variables. BP is exact on trees for discrete variables, where messages are finite vectors of probabilities. For continuous variables the messages become probability *densities* — infinite-dimensional objects — and practical algorithms must close them within a parametric family, almost always the Gaussian one; this is the step that turns BP into AMP, known in physics as the Thouless–Anderson–Palmer (TAP) approach (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016). The classical justification of the Gaussian closure is a central-limit argument valid when every node receives many messages. On a chain each node receives exactly two: the central-limit argument fails by construction, and the quality of the Gaussian closure on chain-structured diffusion posteriors was, to the best of our knowledge, unquantified.

The gap, in one sentence: *there is no end-to-end, exactly solvable account of the joint diffusion score of a structured sequence — its matrix structure, its message-passing computation, and the price of the Gaussian closure when the sequence is not Gaussian.*

1.3 Research questions

The thesis addresses four questions.

RQ1. What is the exact joint score of a Markov chain corrupted by an OU channel, and how does its structure — concretely, the posterior precision matrix — evolve along the diffusion time, from the sparse Markov regime at $t \to 0$ to the isotropic regime at $t \to \infty$?

RQ2. Can the joint score be computed *locally*, by message passing on the chain's factor graph, rather than globally by matrix inversion? In the Gaussian case, how does belief propagation relate to the classical Kalman smoother, and does it reproduce the closed-form score exactly?

RQ3. When the messages are forced into a Gaussian family — the AMP/TAP closure — what survives and what breaks? On the Gaussian chain, is the closure exact; and on a non-Gaussian (Laplace-innovation) chain, how large is the closure error as a function of the diffusion time?

RQ4. What do the exact results imply for *learned* scores: when is a local (banded, convolution-like) score architecture sufficient, and what is the exact law governing the error of local truncations?

1.4 Contributions

The main contributions, each developed in a dedicated chapter, are the following.

1. **A complete closed-form solution of the Gaussian benchmark** (Chapter **??**). For the stationary AR(1) chain under OU corruption we derive the joint score, diagonalise it in closed form, and characterise the full lifecycle of the posterior precision matrix: a corrected *band-fill law* for the small-t growth of its off-diagonals, and exact bulk formulas for its inverse — variance, geometric decay rate $q(\alpha, t)$ of posterior correlations, and their limits. The band-fill law corrects a transcription error in an earlier working note; the correction is validated numerically.

2. **An equation-by-equation message-passing construction of the score** (Chapter **??**). We build the factor graph of the noisy chain, specialise the sum–product equations to it, prove that Gaussian messages are closed under the updates (two elementary facts suffice), and show that the resulting algorithm — Gaussian BP, algebraically identical to a Kalman smoother — reproduces the matrix score to machine precision at cost $O(K)$ instead of $O(K^3)$.

3. **An exact phase boundary for the AMP closure** (Chapter **??**). On the chain, BP and AMP produce the *same score* but *different posterior variances*. We show the difference is a single factor of two in a closure discriminant ($J_d^2 - 4\beta^2$ for BP

against $J_d^2 - 8\beta^2$ for AMP), prove that the BP fixed point exists for every coupling and diffusion time, and derive the exact AMP breakdown time $t_c(\alpha)$ together with the critical coupling $\alpha_c = \sqrt{2} - 1$ below which AMP never breaks down.

4. **A quantitative audit of the Gaussian closure beyond Gaussianity** (Chapters **??** and **??**). For Laplace-innovation chains — where exact inference is still available through deterministic grid quadrature on the same factor graph — we measure the score error of Gaussian-projected BP: it peaks around 20% (relative L^2) at small diffusion time and decays below 1% as the noisy marginals Gaussianise, while the score *direction* remains accurate throughout. We also show that at small t *truncating the inference* (local BP sweeps) is markedly more accurate than *truncating the score matrix* (banding), quantifying the advantage of local message passing over local linear algebra.

5. **A reproducible computational companion.** All results are backed by open, dependency-light Python code with deterministic numerical audits (72/72 and 53/53 independent checks passing); the code and the figures it generates are part of the thesis repository and are listed in Appendix **??**.

1.5 Methodology

The methodology is deliberately old-fashioned: *no neural networks, no black boxes — closed forms first, numerics as audit.* Every claim is either proved by explicit algebra (the derivations are spelled out step by step, with the longer algebraic passages isolated in numbered *toolboxes*) or verified by deterministic numerical checks against brute-force ground truth, in the spirit of the exactly solvable programme of Biroli and Mézard (2023). Where an approximation is introduced (the Gaussian closure of BP messages, local truncations), its error is measured against an exact reference computed by an independent method (matrix algebra or dense grid quadrature). The empirical setting, the audit protocol, and the software packages are described alongside the results and in Appendix **??**.

1.6 Structure of the thesis

The thesis is organised in two movements: a foundations movement (Chapters 2–6) that builds, from first principles, every tool the research movement needs, and a research movement (Chapters ??–??) that contains the original contributions.

Chapter 2 develops statistical mechanics from Hamiltonian mechanics: phase space, Liouville's theorem, the Boltzmann–Gibbs distribution, free energy, and the Ising paradigm of complex systems. Chapter 3 covers stochastic processes and Markov-chain Monte Carlo: stationarity, detailed balance, Metropolis–Hastings and Gibbs sampling, and introduces the AR(1) chain that serves as the running example of the whole thesis. Chapter 4 builds the continuous-time toolkit: Brownian motion, the Itô integral and Itô's lemma, the Ornstein–Uhlenbeck process — the corruption channel of diffusion models — and the Fokker–Planck equation, closing the loop with the Boltzmann distribution of Chapter 2. Chapter 5 treats energy-based models, Markov random fields, hidden Markov models, and factor graphs — the representational language of the research chapters. Chapter 6 assembles these pieces into modern diffusion models: the DDPM construction, the score-SDE formulation, score matching, Tweedie's identity, and the statistical-physics analyses of generative diffusion that motivate this work.

Chapter ?? solves the Gaussian chain exactly (RQ1). Chapter ?? leaves Gaussianity in the most controlled way possible — Laplace innovations — and shows what the joint score looks like when it is no longer linear (first half of RQ3). Chapter ?? recomputes everything by message passing: BP, its Gaussian specialisation and the Kalman connection (RQ2), the AMP closure and its exact phase boundary, the closure error on non-Gaussian chains (RQ3), and the locality laws that answer RQ4. Chapter ?? summarises the findings per research question, discusses limitations, and lays out the research programme that follows from them. Three appendices collect Gaussian identities, the code listings, and the reproducibility protocol.

Chapter 2

From Hamiltonian Mechanics to Statistical Mechanics

Statistical mechanics is the discipline that explains how probabilistic laws emerge from deterministic dynamics. It is also, less obviously, the intellectual ancestor of generative diffusion models: the Boltzmann–Gibbs distribution derived in this chapter is the stationary law of the Langevin dynamics of Chapter 4, the energy functions introduced here become the energy-based models of Chapter 5, and the very idea of *sampling by simulating a physical process* — central to Chapters 3 and 6 — is statistical mechanics turned into an algorithm. We therefore begin at the beginning: with Hamilton's equations. The presentation follows classical references (Goldstein et al., 2002; Landau and Lifshitz, 1980), with the derivations that matter for the rest of the thesis carried out in full.

2.1 Hamiltonian mechanics and phase space

Consider a mechanical system with n degrees of freedom, described by generalised coordinates $q = (q_1, \ldots, q_n)$ and conjugate momenta $p = (p_1, \ldots, p_n)$. Its dynamics is generated by a single scalar function, the *Hamiltonian* $H(q, p)$, which for the systems considered here has the mechanical form

$$H(q,p) \;=\; \underbrace{\sum_{i=1}^{n} \frac{p_i^2}{2m_i}}_{\text{kinetic}} \;+\; \underbrace{U(q_1,\ldots,q_n)}_{\text{potential}}, \tag{2.1}$$

and equals the total energy of the system. The state of the system at any instant is the pair (q,p), a point in the $2n$-dimensional *phase space* $\Gamma \subseteq \mathbb{R}^{2n}$. The evolution is given by *Hamilton's equations*,

$$\dot{q}_i \;=\; \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i \;=\; -\frac{\partial H}{\partial q_i}, \qquad i = 1,\ldots,n. \tag{2.2}$$

Two structural properties of (2.2) are the pillars on which statistical mechanics rests.

Energy conservation. Along any trajectory,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \sum_{i=1}^{n}\left(\frac{\partial H}{\partial q_i}\,\dot{q}_i + \frac{\partial H}{\partial p_i}\,\dot{p}_i\right) = \sum_{i=1}^{n}\left(\frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i}\right) = 0. \tag{2.3}$$

The motion is therefore confined to the *energy shell* $\Gamma_E = \{(q,p) : H(q,p) = E\}$, a $(2n-1)$-dimensional hypersurface.

Incompressibility of the phase flow. The phase-space velocity field

$$v(q,p) \;=\; (\dot{q},\dot{p}) \;=\; \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q}\right) \tag{2.4}$$

has identically vanishing divergence:

$$\nabla\cdot v \;=\; \sum_{i=1}^{n}\left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i}\right) \;=\; \sum_{i=1}^{n}\left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i}\right) \;=\; 0. \tag{2.5}$$

Hamiltonian flow behaves like an incompressible fluid in phase space: it can stretch and fold a region, but never change its volume. This innocuous-looking identity is the seed of the entire probabilistic edifice, as we now show.

2.2 Liouville's theorem

A single trajectory is useless for macroscopic physics: for a gas, $n \sim 10^{23}$ and neither the initial condition nor the solution of (2.2) is accessible. The move that founds statistical mechanics is to describe our *ignorance* of the microstate by a probability density $\rho(q, p, t)$ on phase space: $\rho(q, p, t)\,\mathrm{d}q\,\mathrm{d}p$ is the probability that the system is in the infinitesimal cell around (q, p) at time t. Liouville's theorem states how this density evolves.

Theorem 2.1 (Liouville)**.** *Under the Hamiltonian flow* (2.2)*, the phase-space density obeys*

$$\frac{\partial \rho}{\partial t} \;=\; -\{\rho, H\} \;=\; \sum_{i=1}^{n} \left(\frac{\partial H}{\partial q_i}\frac{\partial \rho}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial \rho}{\partial q_i} \right), \tag{2.6}$$

where $\{\cdot, \cdot\}$ *is the Poisson bracket. Equivalently, the density is constant along trajectories:* $\mathrm{d}\rho/\mathrm{d}t = 0$*.*

Toolbox 2.0 — Proof of Liouville's theorem, step by step

The density is locally conserved — probability mass is neither created nor destroyed, only transported by the flow — so it satisfies the continuity equation familiar from fluid dynamics:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\, v) = 0, \qquad v = (\dot{q}, \dot{p}).$$

Expand the divergence of the product using $\nabla \cdot (\rho v) = v \cdot \nabla \rho + \rho\,(\nabla \cdot v)$:

$$\frac{\partial \rho}{\partial t} + \underbrace{\sum_{i=1}^{n} \left(\dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} \right)}_{v \cdot \nabla \rho} + \rho \underbrace{\sum_{i=1}^{n} \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right)}_{\nabla \cdot v} = 0.$$

By the incompressibility identity (2.5) the last term vanishes. Substituting Hamilton's equations $\dot{q}_i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q_i$ into the middle term:

$$\frac{\partial \rho}{\partial t} + \sum_{i=1}^{n} \left(\frac{\partial H}{\partial p_i}\frac{\partial \rho}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial \rho}{\partial p_i} \right) = 0 \quad \Longleftrightarrow \quad \frac{\partial \rho}{\partial t} = -\{\rho, H\}.$$

Finally, the total (convective) derivative along a trajectory is

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial\rho}{\partial t} + v \cdot \nabla\rho = \frac{\partial\rho}{\partial t} + \{\rho, H\} = 0,$$

which is the second form of the statement: an observer riding a phase-space trajectory sees a constant local density. □

Liouville's theorem has an immediate and profound consequence: *any density that is a function of conserved quantities only is stationary.* If $\rho(q,p) = f\big(H(q,p)\big)$ for some function f, then $\{\rho, H\} = f'(H)\,\{H, H\} = 0$, so $\partial\rho/\partial t = 0$. Equilibrium statistical mechanics is the study of which f nature chooses.

2.3 From dynamics to ensembles

An *ensemble* is a probability distribution over microstates that is stationary under the dynamics and compatible with the macroscopic constraints we impose (fixed energy, fixed temperature, …). Two ensembles matter for this thesis.

2.3.1 The microcanonical ensemble

For a perfectly isolated system, the energy is fixed at E and Liouville's theorem suggests the simplest choice compatible with it: a uniform density on the energy shell,

$$\rho_{\mathrm{mc}}(q,p) \;=\; \frac{1}{\Omega(E)}\,\delta\Big(H(q,p) - E\Big), \qquad \Omega(E) = \int_\Gamma \delta\Big(H(q,p) - E\Big)\,\mathrm{d}q\,\mathrm{d}p, \tag{2.7}$$

where $\Omega(E)$ counts the microstates at energy E. The postulate that all accessible microstates are equally likely — the *equal a priori probability postulate* — is motivated (not proved) by ergodicity: for ergodic dynamics, time averages along a single trajectory converge to averages over the shell, so the uniform measure is the one a long experiment effectively samples. Boltzmann's entropy attaches a number to this count,

$$S(E) \;=\; k_B \log \Omega(E), \tag{2.8}$$

with k_B the Boltzmann constant. Entropy is the logarithm of the number of microscopic ways a macroscopic state can be realised.

2.3.2 The canonical ensemble and the Boltzmann distribution

Isolated systems are an idealisation; the physically ubiquitous situation is a system exchanging energy with a much larger environment (a *heat bath*) at temperature T. The stationary distribution of the system alone is then no longer uniform: it is the *Boltzmann–Gibbs distribution*,

$$\boxed{\rho_\beta(x) \;=\; \frac{1}{Z(\beta)}\, e^{-\beta H(x)}, \qquad Z(\beta) = \int e^{-\beta H(x)}\, \mathrm{d}x, \qquad \beta = \frac{1}{k_B T},} \tag{2.9}$$

where $x = (q, p)$ collects the microstate and $Z(\beta)$, the *partition function*, normalises the density. Equation (2.9) can be derived in two independent ways, and both derivations are instructive enough to be carried out in full: the first (physical) traces the exponential to the entropy of the bath; the second (information-theoretic) characterises it as the least-committal distribution compatible with a fixed mean energy, a perspective due to Jaynes (1957) that returns, almost verbatim, in modern machine learning.

Toolbox 2.1 — The Boltzmann distribution from the heat bath

Let the total system = (system S) $\cup$ (bath B) be isolated with total energy E_{tot}, and let the coupling be weak, so $H_{\text{tot}} = H_S + H_B$. Apply the microcanonical postulate to the total system: the probability that S is in a *specific* microstate x with energy $H_S(x) = \epsilon$ is proportional to the number of bath microstates carrying the remaining energy,

$$\mathbb{P}(x) \;\propto\; \Omega_B\Big(E_{\text{tot}} - \epsilon\Big) \;=\; \exp\Big[\tfrac{1}{k_B} S_B\Big(E_{\text{tot}} - \epsilon\Big)\Big]\,,$$

using the definition (2.8) of the bath entropy. Since the bath is huge, $\epsilon \ll E_{\mathrm{tot}}$, expand S_B to first order:

$$S_B(E_{\mathrm{tot}} - \epsilon) = S_B(E_{\mathrm{tot}}) - \epsilon \left.\frac{\partial S_B}{\partial E}\right|_{E_{\mathrm{tot}}} + O\Big(\epsilon^2/E_{\mathrm{tot}}\Big).$$

The derivative of the bath entropy with respect to its energy *defines* the bath temperature, $\partial S_B/\partial E = 1/T$. Substituting,

$$\mathbb{P}(x) \;\propto\; \underbrace{e^{S_B(E_{\mathrm{tot}})/k_B}}_{\text{constant in } x} \cdot e^{-\epsilon/(k_B T)} \;\propto\; e^{-\beta H_S(x)}.$$

The quadratic remainder is suppressed by the bath's heat capacity and vanishes in the thermodynamic limit. Normalising yields (2.9). Note what happened: the exponential form is nothing but the first-order Taylor expansion of the bath's entropy — the Boltzmann factor is the price, in bath microstates, of lending energy ϵ to the system. $\square$

Toolbox 2.2 — The Boltzmann distribution from maximum entropy

Forget physics; ask, with Jaynes (1957): among all densities ρ with a given mean energy $\mathbb{E}_\rho[H] = U$, which one assumes the least beyond the constraint? "Least" is measured by the Gibbs–Shannon entropy $S[\rho] = -k_B \int \rho \log \rho$. Maximise $S[\rho]$ subject to $\int \rho = 1$ and $\int \rho H = U$ with Lagrange multipliers λ_0, λ_1:

$$\mathcal{L}[\rho] = -k_B \int \rho \log \rho \, \mathrm{d}x \;-\; \lambda_0\Big(\int \rho \, \mathrm{d}x - 1\Big) \;-\; \lambda_1\Big(\int \rho H \, \mathrm{d}x - U\Big).$$

Take the functional derivative and set it to zero pointwise:

$$\frac{\delta \mathcal{L}}{\delta \rho(x)} = -k_B\Big(\log \rho(x) + 1\Big) - \lambda_0 - \lambda_1 H(x) = 0 \;\Longrightarrow\; \rho(x) = \exp\Big[-1 - \tfrac{\lambda_0}{k_B} - \tfrac{\lambda_1}{k_B} H(x)\Big].$$

The density is an exponential of the constrained quantity — this is completely general: constraining $\mathbb{E}[\phi_j(x)]$ for features ϕ_j produces $\rho \propto \exp[-\sum_j \beta_j \phi_j(x)]$, the exponential family. The multiplier λ_0 is fixed by normalisation and produces Z; renaming $\lambda_1/k_B = \beta$ and identifying $\beta = 1/(k_B T)$ through thermodynamics recovers (2.9) exactly. Second-order check: $\delta^2\mathcal{L}/\delta\rho^2 = -k_B/\rho < 0$, so the stationary point is the unique maximum (the entropy is strictly concave). □

The second derivation deserves a remark that will echo through the thesis: the Boltzmann distribution is not a statement about molecules but about *inference under constraints*. Any time we posit a model of the form $p(x) \propto e^{-E(x)}$ — as we will for Markov random fields and energy-based models in Chapter 5 — we are doing statistical mechanics, whether the variables are spins, pixels, or the frames of a video.

2.4 The partition function and free energy

The partition function $Z(\beta)$ in (2.9) looks like a mere normaliser; it is in fact the central computational object of the theory — and the reason approximate inference exists as a field. Its logarithm defines the *Helmholtz free energy*,

$$F(\beta) \;=\; -\frac{1}{\beta}\log Z(\beta). \tag{2.10}$$

Toolbox 2.3 — Everything is a derivative of $\log Z$

Differentiate $\log Z$ with respect to β, carrying the derivative through the integral:

$$-\frac{\partial \log Z}{\partial \beta} = -\frac{1}{Z}\frac{\partial}{\partial \beta}\int e^{-\beta H} = \frac{\int H\, e^{-\beta H}}{\int e^{-\beta H}} = \mathbb{E}_{\rho_\beta}[H] \;=\; U,$$

the mean energy. Differentiate once more:

$$\begin{aligned}\frac{\partial^2 \log Z}{\partial \beta^2} = \frac{\partial}{\partial \beta}\left(-\frac{\int He^{-\beta H}}{Z}\right) &= \frac{\int H^2 e^{-\beta H}}{Z} - \left(\frac{\int He^{-\beta H}}{Z}\right)^2 \\ &= \mathbb{E}[H^2] - \mathbb{E}[H]^2 \;=\; \mathrm{Var}(H) \;\geq\; 0.\end{aligned}$$

So $\log Z$ is convex in β, its first derivative gives the mean energy, its second the energy fluctuations; the latter equals $k_B T^2 C_V$ with C_V the heat capacity — a *fluctuation–dissipation* relation: how a system responds to heating is determined by its spontaneous equilibrium fluctuations. Finally, rewriting the Gibbs–Shannon entropy of ρ_β:

$$S[\rho_\beta] = -k_B \mathbb{E}\Big[\log \rho_\beta\Big] = -k_B \mathbb{E}\Big[-\beta H - \log Z\Big] = \frac{U}{T} + k_B \log Z,$$

and multiplying by $-T$: $-TS = -U + F$, i.e.

$$F = U - TS.$$

The free energy is the exact trade-off between energy and entropy: at low T minimising F means minimising energy (ordered states win), at high T it means maximising entropy (disordered states win). Phase transitions are the non-analyticities of F where the winner changes. $\square$

There is one more characterisation of F worth recording, because it is the statistical-mechanics form of the variational principle that underlies both the ELBO of diffusion models (Chapter 6) and the Bethe/BP free energies (Chapter **??**). For *any* trial density μ,

$$\underbrace{\mathbb{E}_\mu[H] - \tfrac{1}{\beta} S[\mu]/k_B}_{\text{variational free energy } F[\mu]} \;=\; F \;+\; \tfrac{1}{\beta}\, D_{\mathrm{KL}}(\mu \,\|\, \rho_\beta) \;\geq\; F, \tag{2.11}$$

with equality iff $\mu = \rho_\beta$: the Boltzmann distribution is the unique minimiser of the variational free energy, and the gap is exactly a Kullback–Leibler divergence. Every variational method in machine learning is a restriction of (2.11) to a tractable family of μ's.

2.5 Interacting systems: the Ising paradigm

Everything so far holds for arbitrary H; the physics (and the difficulty) enters when the energy couples many degrees of freedom. The canonical example is the *Ising model* (Ising, 1925): binary spins $\sigma_i \in \{-1, +1\}$ on the nodes of a graph $G = (V, \mathcal{E})$, with energy

$$H(\sigma) \;=\; -\sum_{(i,j)\in\mathcal{E}} J_{ij}\,\sigma_i\sigma_j \;-\; \sum_{i\in V} h_i\,\sigma_i, \tag{2.12}$$

where J_{ij} are pairwise couplings and h_i external fields. The Gibbs distribution $p(\sigma) \propto e^{-\beta H(\sigma)}$ makes neighbouring spins prefer alignment when $J_{ij} > 0$. Three lessons from this model shape the rest of the thesis.

(i) The partition function is the hard part. $Z = \sum_\sigma e^{-\beta H(\sigma)}$ runs over $2^{|V|}$ configurations. On two-dimensional lattices Onsager computed it exactly (Onsager, 1944); on general graphs the computation is #P-hard, and *all* of approximate inference — MCMC (Chapter 3), variational methods, message passing (Chapter **??**) — exists because of this wall. On *trees*, however, Z and all marginals are computable exactly in linear time by the transfer-matrix/belief-propagation recursion; this tractable island is precisely where our research model lives.

(ii) Collective order emerges. For β beyond a critical value the two-dimensional model magnetises spontaneously: a macroscopic fraction of spins align even at $h = 0$, and the symmetric state splits into two ergodic components. This spontaneous symmetry breaking is the archetype of a *phase transition*, and its dynamical counterpart — a generation trajectory committing to one mode of a multimodal distribution — is exactly the *speciation* phenomenon of diffusion models (Biroli et al., 2024; Raya and Ambrogioni, 2023).

(iii) Disorder creates complexity. Making the couplings J_{ij} random (of both signs) yields spin glasses (Edwards and Anderson, 1975; Sherrington and Kirkpatrick, 1975): frustrated systems whose free-energy landscape shatters into exponentially many metastable valleys. The mean-field theory of these systems (Mézard et al., 1987) produced the cavity method and the TAP equations (Thouless et al., 1977) — the direct ancestors of the message-passing algorithms we deploy in Chapter **??** — and its modern export to inference problems (Zdeborová and Krzakala, 2016; Mézard and Montanari, 2009) is the methodological backbone of this thesis.

2.6 Complex systems, and what this thesis takes from physics

A *complex system* is, operationally, a system whose macroscopic behaviour is not readable off its microscopic rules: interactions generate collective phenomena — order, phases, transitions, slow dynamics — that must be derived, not postulated. Statistical mechanics is the toolbox for that derivation, and its objects map one-to-one onto the objects of modern generative modelling:

Statistical mechanics	**Generative modelling**
microstate x	data point / latent configuration
energy $H(x)$	negative log-density $-\log p(x)$ (up to Z)
Boltzmann distribution $e^{-\beta H}/Z$	model distribution
partition function Z	normalising constant / evidence
free energy F	negative ELBO (up to constants)
Langevin dynamics	sampling / reverse diffusion
phase transition	speciation, memorisation onset
cavity / TAP equations	belief propagation / AMP

The next four chapters make each row of this dictionary precise, starting with the probabilistic dynamics — Markov chains and their continuous-time limits — that turn the static ensembles of this chapter into processes that *move* probability around, which is what a diffusion model ultimately is.

Chapter 3

Stochastic Processes and Markov Chain Monte Carlo

Chapter 2 answered the question *"which distribution?"* — Boltzmann's. This chapter answers *"how do we reach it, and how do we move within it?"*. The answer is the theory of Markov processes: chains whose future depends on the past only through the present. Markov chains play a triple role in this thesis: they are the *algorithms* that sample from intractable distributions (MCMC), they are the *data* whose diffusion score we compute exactly (the AR(1) sequence introduced in Section 3.2), and — run in continuous time — they are the *corruption and generation processes* of diffusion models (Chapters 4 and 6). Standard references for the material are Robert and Casella (2004) and Brockwell and Davis (1991).

3.1 Markov chains: kernels, stationarity, reversibility

Definition 3.1 (Markov chain)**.** A sequence of random variables $(X_k)_{k\geq 0}$ with values in a state space $\mathcal{X}$ is a *Markov chain* if for every k and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}\Big(X_{k+1} \in A \,\Big|\, X_k, X_{k-1}, \ldots, X_0\Big) \;=\; \mathbb{P}\Big(X_{k+1} \in A \,\Big|\, X_k\Big). \tag{3.1}$$

The conditional law is encoded by a *transition kernel* $M(x' \mid x)$: a density (or matrix, for discrete $\mathcal{X}$) with $\int M(x'|x)\,\mathrm{d}x' = 1$ for every x.

The Markov property (3.1) is exactly a statement about *factorisation*: the joint law of the first K states is

$$p(x_0, x_1, \ldots, x_{K-1}) \;=\; p_0(x_0) \prod_{k=0}^{K-2} M(x_{k+1} \mid x_k). \tag{3.2}$$

This one-line factorisation is the single most important equation of the thesis. It says that the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables — the structure that makes the chain a *tree* in the factor-graph sense of Chapter 5, that makes its precision matrix tridiagonal in the Gaussian case of Chapter **??**, and that makes belief propagation exact in Chapter **??**.

Marginal distributions evolve by integrating the kernel: $p_{k+1}(x') = \int M(x'|x)\, p_k(x)\,\mathrm{d}x$, compactly $p_{k+1} = Mp_k$.

Definition 3.2 (Stationarity)**.** A distribution π is *stationary* (or invariant) for M if $\pi = M\pi$, i.e.

$$\pi(x') \;=\; \int M(x' \mid x)\, \pi(x)\,\mathrm{d}x \qquad \text{for all } x'. \tag{3.3}$$

A chain started in π stays in π forever; the trajectory $(X_0, X_1, \ldots)$ is then a *stationary process*, and all its finite-dimensional laws are invariant under time shifts.

Definition 3.3 (Detailed balance / reversibility)**.** The kernel M satisfies *detailed balance* with respect to π if

$$\pi(x)\, M(x' \,|\, x) \;=\; \pi(x')\, M(x \,|\, x') \qquad \text{for all } x, x'. \tag{3.4}$$

Detailed balance says the equilibrium probability flux from x to x' equals the reverse flux — the chain, watched in equilibrium, is statistically indistinguishable run forwards or backwards. It is a *sufficient* condition for stationarity: integrating (3.4) over x gives $\int \pi(x) M(x'|x)\,\mathrm{d}x \;=\; \pi(x') \int M(x|x')\,\mathrm{d}x \;=\; \pi(x')$, which is (3.3). Its power is practical: detailed balance is a *local, pointwise* condition, checkable without ever computing an integral — which is why every MCMC algorithm in Section 3.3 is engineered to satisfy it.

Stationarity alone does not guarantee that the chain *finds* π; for that one needs the chain to be able to reach everywhere (*irreducibility*) without getting trapped in cycles (*aperiodicity*). Under these conditions the fundamental convergence theorem holds (Robert and Casella, 2004): the marginal law of X_k converges to π from any initial condition, and time averages converge to expectations under π (the ergodic theorem),

$$\frac{1}{N} \sum_{k=1}^{N} f(X_k) \;\xrightarrow[N\to\infty]{\text{a.s.}}\; \mathbb{E}_\pi[f]. \tag{3.5}$$

Equation (3.5) is the rigorous form of the ergodic postulate invoked in Section 2.3, with the physical dynamics replaced by an algorithmic one.

3.2 The AR(1) chain: the running example of the thesis

We now introduce the specific Markov chain that the research chapters solve completely: the first-order autoregressive (AR(1)) process, the canonical Gaussian Markov chain (Brockwell and Davis, 1991). Let $\alpha \in (-1, 1)$ and let (η_k) be i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$ innovations; define

$$a_{k+1} \;=\; \alpha\, a_k + \eta_k, \qquad k = 0, 1, \ldots, K-2. \tag{3.6}$$

The transition kernel is Gaussian, $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, and the joint law of the trajectory follows from (3.2). Everything about this chain is computable in closed form; the next toolbox derives the three facts used throughout.

Toolbox 3.0 — Stationary distribution and autocovariance of AR(1)

(i) Stationary variance. Suppose $a_k \sim \mathcal{N}(0, \sigma^2)$ and impose that a_{k+1} has the same variance. Since η_k is independent of a_k,

$$\mathrm{Var}(a_{k+1}) = \alpha^2\,\mathrm{Var}(a_k) + \sigma_\eta^2 \stackrel{!}{=} \mathrm{Var}(a_k) \quad\Longrightarrow\quad \sigma^2 = \alpha^2\sigma^2 + \sigma_\eta^2 \quad\Longrightarrow\quad \sigma_\infty^2 = \frac{\sigma_\eta^2}{1-\alpha^2}.$$

The fixed point exists precisely because $|\alpha| < 1$; the map $\sigma^2 \mapsto \alpha^2\sigma^2 + \sigma_\eta^2$ is a contraction, so *any* initial variance converges to σ_∞^2 geometrically. Since Gaussianity is preserved by the affine recursion, the stationary law is $\pi = \mathcal{N}(0, \sigma_\infty^2)$.

(ii) Autocovariance. In the stationary regime, for $d \geq 1$, unroll the recursion d times: $a_{k+d} = \alpha^d a_k + \sum_{j=0}^{d-1} \alpha^j \eta_{k+d-1-j}$. All innovation terms are independent of a_k, hence

$$\mathrm{Cov}(a_k, a_{k+d}) = \alpha^d\,\mathrm{Var}(a_k) = \alpha^d\,\sigma_\infty^2,$$

and by symmetry $\mathrm{Cov}(a_i, a_j) = \alpha^{|i-j|}\sigma_\infty^2$ for all i, j: correlations decay *geometrically* with the lag, with rate α. The covariance matrix of K consecutive frames,

$$(\Sigma_0)_{ij} = \sigma_\infty^2\,\alpha^{|i-j|},$$

depends on (i, j) only through $i - j$: it is a symmetric *Toeplitz* matrix, a structure we exploit heavily in Chapter **??**.

(iii) Detailed balance. With $\pi = \mathcal{N}(0, \sigma_\infty^2)$ and $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, both sides of (3.4) are proportional to

$$\exp\left[-\frac{a^2}{2\sigma_\infty^2} - \frac{(a' - \alpha a)^2}{2\sigma_\eta^2}\right] = \exp\left[-\frac{(1-\alpha^2)a^2 + (a')^2 - 2\alpha a a' + \alpha^2 a^2}{2\sigma_\eta^2}\right],$$

and the exponent $-\Big[a^2 + (a')^2 - 2\alpha a a'\Big]/(2\sigma_\eta^2)$ is *symmetric* in (a, a'). Hence the stationary AR(1) chain is reversible: statistically, the movie of the chain looks the same played in either direction. $\square$

The AR(1) chain is thus the perfect miniature of everything this chapter discussed: a Markov kernel with an explicit stationary law, geometric convergence, reversibility, and — decisive for us — *exact Gaussian algebra* at every step. When in Chapter **??** we corrupt each frame of this chain with independent noise and ask for the joint score of the result, every quantity will remain in closed form.

3.3 Markov chain Monte Carlo

MCMC inverts the logic of the previous section. There, a kernel M was given and we asked for its stationary law π. In sampling problems we are given π — typically a Boltzmann distribution $\pi(x) \propto e^{-\beta H(x)}$ known only up to its partition function — and we must *design* a kernel M whose stationary law is π. The ergodic theorem (3.5) then converts a single long trajectory into expectation values. The idea originates, fittingly, in computational statistical mechanics (Metropolis et al., 1953).

3.3.1 The Metropolis–Hastings algorithm

Let $g(x'|x)$ be any proposal kernel we can sample from. The Metropolis–Hastings (MH) chain (Metropolis et al., 1953; Hastings, 1970) moves from x by (i) proposing $x' \sim g(\cdot|x)$, (ii) accepting the move with probability

$$A(x \to x') \;=\; \min\left\{1,\ \frac{\pi(x')\, g(x \,|\, x')}{\pi(x)\, g(x' \,|\, x)}\right\}, \tag{3.7}$$

and (iii) staying at x otherwise. Two features make (3.7) remarkable. First, π enters only through the *ratio* $\pi(x')/\pi(x)$: the intractable partition function cancels, so MH samples from $e^{-\beta H}/Z$ while only ever evaluating energy differences. Second, the rule is engineered to satisfy detailed balance:

Toolbox 3.1 — Detailed balance for Metropolis–Hastings

The effective transition density for an accepted move $x \neq x'$ is $M(x'|x) = g(x'|x)\,A(x \to x')$. Compute the flux from x to x':

$$\pi(x)\,M(x'|x) = \pi(x)\,g(x'|x)\,\min\left\{1, \frac{\pi(x')\,g(x|x')}{\pi(x)\,g(x'|x)}\right\} = \min\left\{\pi(x)\,g(x'|x),\ \pi(x')\,g(x|x')\right\},$$

where the second equality uses $c\min\{1, b/c\} = \min\{c, b\}$ for $c > 0$. The final expression is *symmetric* under $x \leftrightarrow x'$; therefore

$$\pi(x)\,M(x'|x) = \pi(x')\,M(x|x'),$$

which is detailed balance (3.4). The rejection branch (staying at x) contributes only to the diagonal of the kernel and cannot violate the balance of fluxes between distinct states. Stationarity of π follows by the integration argument of Section 3.1; with an irreducible, aperiodic proposal, the convergence theorem applies. □

3.3.2 Gibbs sampling

When $x = (x_1, \ldots, x_n)$ is high-dimensional but each *conditional* $\pi(x_i \mid x_{\setminus i})$ is tractable, the Gibbs sampler (Geman and Geman, 1984) cycles through the coordinates, resampling each from its conditional given the current values of the others. Each coordinate update is an MH move with acceptance probability identically 1, so detailed balance holds coordinate-wise and π is stationary for the composite kernel. The Gibbs sampler thrives exactly when the model has sparse conditional structure — for a Markov chain, $\pi(x_k|x_{\setminus k})$ depends only on the neighbours (x_{k-1}, x_{k+1}): locality of the model makes the algorithm local. The reader should hear, already, the theme of Chapter **??**: *inference cost follows graph structure.*

3.3.3 Langevin dynamics: MCMC without rejections

For continuous x and differentiable energy, a different design principle uses gradients. Consider the discrete-time update

$$x_{k+1} \;=\; x_k + \epsilon\,\nabla \log \pi(x_k) \;+\; \sqrt{2\epsilon}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I), \tag{3.8}$$

the (unadjusted) *Langevin algorithm*. The drift pushes towards high probability; the noise maintains exploration. Two observations preview the next chapters. First, (3.8) only requires the *score* $\nabla \log \pi$ — the normalising constant is never needed; this is precisely why the score is the natural object for generative modelling (Chapter 6), and why so much of this thesis is devoted to computing scores exactly. Second, as $\epsilon \to 0$ the recursion (3.8) becomes a stochastic differential equation — the Langevin diffusion — whose stationary law is exactly π; making that limit rigorous requires the Itô calculus and Fokker–Planck theory of Chapter 4, where we will verify the claim by direct computation.

3.4 From algorithmic chains to data chains

A closing change of perspective, to set up the research problem. In this chapter Markov chains were *algorithms*: their stationary law was the target, their trajectory a means to an end. From Chapter **??** onwards the chain (3.6) plays the opposite role: it is the *data-generating process*, its trajectory $(a_0, \ldots, a_{K-1})$ is one data point — a "video" of K frames — and the object of interest is the probability law of the *whole trajectory* after each frame has been independently corrupted by noise. The factorisation (3.2) then stops being a computational convenience and becomes the *structure to be discovered*: the entire research programme of this thesis can be phrased as asking how much of (3.2) survives noising, and how an inference algorithm — or a neural network — can exploit what survives.

Chapter 4

Stochastic Differential Equations, Itô Calculus, and the Fokker–Planck Picture

Diffusion models corrupt data by running a stochastic differential equation (SDE) and generate data by running another SDE backwards. This chapter builds that machinery from the ground up: Brownian motion, the Itô integral and Itô's lemma, the Ornstein–Uhlenbeck (OU) process — the corruption channel used throughout the thesis — and the Fokker–Planck equation, which transports the *density* rather than the sample and closes the circle opened in Chapter 2: the stationary law of Langevin dynamics is exactly the Boltzmann distribution. References: Øksendal (2003); Gardiner (2009); Risken (1996).

4.1 From ODEs to SDEs

An ordinary differential equation $\dot{x} = f(x, t)$ moves a point along a deterministic velocity field. Physical and financial systems, however, feel rapid, unresolved forces — collisions, order flow, thermal agitation — whose cumulative effect is well modelled by adding noise:

$$\mathrm{d}X_t \;=\; \underbrace{f(X_t,t)\,\mathrm{d}t}_{\text{drift}} \;+\; \underbrace{g(t)\,\mathrm{d}W_t}_{\text{diffusion}}. \tag{4.1}$$

Here W_t is a *Wiener process* (standard Brownian motion): the unique process with $W_0 = 0$, independent increments, $W_t - W_s \sim \mathcal{N}(0, t-s)$, and continuous paths. Equation (4.1) is shorthand for the integral equation $X_t = X_0 + \int_0^t f(X_s,s)\,\mathrm{d}s + \int_0^t g(s)\,\mathrm{d}W_s$, and the whole subtlety lives in the last term: Brownian paths are nowhere differentiable, so $\int g\,\mathrm{d}W$ cannot be a Riemann–Stieltjes integral. Itô's construction (Itô, 1944) defines it as the limit of left-endpoint sums,

$$\int_0^t g(s)\,\mathrm{d}W_s \;=\; \lim_{n\to\infty} \sum_{j=0}^{n-1} g(s_j)\left(W_{s_{j+1}} - W_{s_j}\right), \tag{4.2}$$

where the evaluation at the *left* endpoint s_j makes the integrand independent of the coming increment. Two consequences follow immediately: the Itô integral is a martingale (zero mean), and its variance is given by the *Itô isometry*, $\mathbb{E}\left[(\int_0^t g\,\mathrm{d}W)^2\right] = \int_0^t g(s)^2\,\mathrm{d}s$, both because cross terms $\mathbb{E}[g(s_j)(W_{s_{j+1}} - W_{s_j})]$ vanish by independence.

The price of the construction is a new chain rule. Its origin is the *quadratic variation* of Brownian motion: increments scale as $\mathrm{d}W_t \sim \sqrt{\mathrm{d}t}$, so second-order terms in $\mathrm{d}W$ are first-order in $\mathrm{d}t$ and cannot be discarded.

Toolbox 4.0 — Itô's lemma, and where the extra term comes from

Let $Y_t = \phi(X_t,t)$ with ϕ twice differentiable and X_t solving (4.1). Taylor-expand ϕ to second order:

$$\mathrm{d}Y = \frac{\partial\phi}{\partial t}\,\mathrm{d}t + \frac{\partial\phi}{\partial x}\,\mathrm{d}X + \frac{1}{2}\frac{\partial^2\phi}{\partial x^2}\,(\mathrm{d}X)^2 + \dots$$

Substitute $\mathrm{d}X = f\,\mathrm{d}t + g\,\mathrm{d}W$ and expand the square:

$$(\mathrm{d}X)^2 = f^2(\mathrm{d}t)^2 + 2fg\,\mathrm{d}t\,\mathrm{d}W + g^2(\mathrm{d}W)^2.$$

Now use the multiplication table of Itô calculus, which is nothing but the scaling

$\mathrm{d}W \sim \sqrt{\mathrm{d}t}$ made systematic:

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\,\mathrm{d}W = 0, \qquad (\mathrm{d}W)^2 = \mathrm{d}t.$$

The last identity is the rigorous statement that the quadratic variation of W over $[0,t]$ equals t: for a partition of mesh $\to 0$, $\sum_j (W_{s_{j+1}} - W_{s_j})^2 \to t$ in mean square, since the sum has mean $\sum_j (s_{j+1} - s_j) = t$ and variance $2\sum_j (s_{j+1} - s_j)^2 \to 0$. Keeping the surviving terms:

$$\mathrm{d}Y = \left(\frac{\partial\phi}{\partial t} + f\,\frac{\partial\phi}{\partial x} + \frac{g^2}{2}\,\frac{\partial^2\phi}{\partial x^2}\right)\mathrm{d}t \;+\; g\,\frac{\partial\phi}{\partial x}\,\mathrm{d}W.$$

The half-Laplacian term — absent from the classical chain rule — is the engine of everything that follows: it is why densities *spread*, why the Fokker–Planck equation has a second-order term, and why exp-transformations of SDEs pick up variance corrections. $\square$

4.2 The Ornstein–Uhlenbeck process: the corruption channel

The noising process used by diffusion models — and by every result of this thesis — is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled back to the origin by a linear spring,

$$\mathrm{d}X_t \;=\; -X_t\,\mathrm{d}t \;+\; \sqrt{2}\,\mathrm{d}W_t, \qquad X_0 = a. \tag{4.3}$$

(The drift and diffusion constants are a normalisation choice — this is the "variance-preserving" convention of the diffusion literature (Song et al., 2021); any other choice rescales time.) The OU process is the unique Gaussian, Markov, stationary process in continuous time, and it is exactly solvable.

Toolbox 4.1 — Exact solution of the OU SDE by integrating factor

Multiply (4.3) by e^t and consider $Y_t = e^t X_t$. By Itô's lemma with $\phi(x,t) = e^t x$ (here $\partial^2\phi/\partial x^2 = 0$, so no correction term arises):

$$\mathrm{d}Y_t = e^t X_t\,\mathrm{d}t + e^t\,\mathrm{d}X_t = e^t X_t\,\mathrm{d}t + e^t\Big(-X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t\Big) = \sqrt{2}\,e^t\,\mathrm{d}W_t.$$

The drift cancels exactly — that is what the integrating factor is for. Integrate from 0 to t and undo the substitution:

$$X_t \;=\; a\,e^{-t} \;+\; \sqrt{2}\int_0^t e^{-(t-s)}\,\mathrm{d}W_s.$$

The integral is Gaussian (limit of Gaussian sums), with mean zero and variance given by the Itô isometry:

$$\mathrm{Var}(X_t) = 2\int_0^t e^{-2(t-s)}\,\mathrm{d}s = 2\,e^{-2t}\int_0^t e^{2s}\,\mathrm{d}s = 2\,e^{-2t}\,\frac{e^{2t}-1}{2} = 1 - e^{-2t}.$$

Hence the *transition kernel* of the OU channel is

$$p(x_t \,|\, X_0 = a) \;=\; \mathcal{N}\Big(x_t;\; a\,e^{-t},\; \Delta_t\Big), \qquad \boxed{\Delta_t \;=\; 1 - e^{-2t}}.$$

□

The boxed kernel is used on every page of Chapters **??** –**??**, so we fix its reading now. Corrupting a data point a for a time t produces

$$x \;=\; a\,e^{-t} \;+\; \sqrt{\Delta_t}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1) : \tag{4.4}$$

the signal is *contracted* by e^{-t} while isotropic noise of variance Δ_t fills in. At $t = 0$: $x = a$ (no corruption); as $t \to \infty$: $x \sim \mathcal{N}(0,1)$ regardless of a (all information destroyed, the equilibrium of the spring). The signal-to-noise ratio e^{-2t}/Δ_t sweeps continuously from ∞ to 0: the OU channel interpolates between the data distribution and a standard Gaussian, which is precisely what a diffusion model needs.

Two structural remarks, both load-bearing later. First, the OU kernel is Gaussian *in a as well as in x*: viewed as a function of the clean variable, $\mathcal{N}(x; ae^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - xe^t)^2]$ — a Gaussian "likelihood" factor. This is the fact that makes the posteriors of Chapter **??** jointly Gaussian and the messages of Chapter **??** closable. Second, if a random vector $a \sim \mathcal{N}(\mu, \Sigma_0)$ is pushed through (4.4) coordinate-wise with *independent* noises, the output is Gaussian with

$$\mu_t = e^{-t}\mu, \qquad \Sigma_t \;=\; e^{-2t}\,\Sigma_0 \;+\; \Delta_t\, I, \tag{4.5}$$

since means and covariances transform linearly and the added noise is white. Equation (4.5) — shrink the covariance, add a multiple of the identity — is the entire "forward process" of this thesis in one line.

4.3 The Fokker–Planck equation: transporting densities

The SDE (4.1) moves *samples*; the corresponding motion of the *density* $p_t(x)$ is governed by the Fokker–Planck (or forward Kolmogorov) equation. This change of viewpoint — from trajectories to densities — is the same one Liouville's theorem performed for Hamiltonian flow, and the derivation is parallel, with the Itô correction supplying a new, diffusive term.

Toolbox 4.2 — Derivation of the Fokker–Planck equation

Let ϕ be an arbitrary smooth test function of compact support, and compute $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\phi(X_t)]$ in two ways.

Way 1: through the samples. By Itô's lemma,

$$\mathrm{d}\phi(X_t) = \Big(f\,\phi' + \tfrac{g^2}{2}\,\phi''\Big)\mathrm{d}t + g\,\phi'\,\mathrm{d}W_t,$$

and taking expectations kills the martingale term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}[\phi(X_t)] = \mathbb{E}\Big[f(X_t,t)\,\phi'(X_t)\Big] + \frac{g(t)^2}{2}\,\mathbb{E}\Big[\phi''(X_t)\Big].$$

Way 2: through the density. $\mathbb{E}[\phi(X_t)] = \int \phi(x)\,p_t(x)\,\mathrm{d}x$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}[\phi(X_t)] = \int \phi(x)\,\frac{\partial p_t(x)}{\partial t}\,\mathrm{d}x.$$

Equate the two, and integrate Way 1 by parts to move all derivatives off ϕ (boundary terms vanish by compact support): once for the drift term, twice for the diffusion term,

$$\int \phi\,\frac{\partial p_t}{\partial t} = \int \phi\left[-\frac{\partial}{\partial x}\big(f\,p_t\big) + \frac{g^2}{2}\,\frac{\partial^2 p_t}{\partial x^2}\right]\mathrm{d}x.$$

Since ϕ is arbitrary, the integrands must agree pointwise:

$$\boxed{\frac{\partial p_t}{\partial t} = -\frac{\partial}{\partial x}\Big(f(x,t)\,p_t(x)\Big) + \frac{g(t)^2}{2}\,\frac{\partial^2 p_t(x)}{\partial x^2}.}$$

The first term transports probability along the drift (as in Liouville); the second spreads it diffusively — it is the macroscopic footprint of the Itô correction. □

4.3.1 Stationary law of Langevin dynamics: closing the loop

Apply the Fokker–Planck equation to the *Langevin SDE* associated with a potential U (the continuous-time limit of the algorithm (3.8) of Chapter 3):

$$\mathrm{d}X_t = -\nabla U(X_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t. \tag{4.6}$$

A stationary density solves $0 = \nabla\cdot(\nabla U\,p + \beta^{-1}\nabla p)$, i.e. the probability current $J = -(\nabla U)\,p - \beta^{-1}\nabla p$ is divergence-free; imposing the natural (decaying) boundary conditions forces $J \equiv 0$, hence $\nabla p/p = -\beta\nabla U$, hence

$$p_\infty(x) \;\propto\; e^{-\beta U(x)}\;: \tag{4.7}$$

Langevin dynamics equilibrates to the Boltzmann distribution of Chapter 2. This is the two-line proof of the claim left pending in Section 3.3, and it explains the deep kinship of sampling and physics: a Langevin sampler *is* an overdamped physical system at temperature $1/\beta$. Note also that the drift of (4.6) is (up to β) the *score* of the target, $-\nabla U = \nabla \log p_\infty$: dynamics that sample a distribution are driven by its score.

For the OU process, $U(x) = x^2/2$ and $\beta = 1$: the stationary law is $\mathcal{N}(0, 1)$, confirming the $t \to \infty$ limit of Section 4.2.

4.4 Time reversal: the mother theorem of diffusion models

One last piece of SDE theory converts everything above into a generative method. Suppose data $X_0 \sim p_0$ are corrupted by a forward SDE (4.1) up to time T, producing marginals p_t. Can the corruption be undone — can we find an SDE which, run from $X_T \sim p_T$ backwards, reproduces the same marginals? The answer (Anderson, 1982) is yes, and the reverse drift needs exactly one unknown object:

$$\mathrm{d}X_t \;=\; \Big[f(X_t, t) \;-\; g(t)^2\, \nabla_x \log p_t(X_t)\Big]\,\mathrm{d}t \;+\; g(t)\,\mathrm{d}\bar{W}_t, \tag{4.8}$$

where time runs backwards from T to 0 and $\bar{W}$ is a reverse-time Wiener process. The correction to the drift is the *score* $\nabla_x \log p_t$ of the noised marginal — the object defined in (1.1) and computed exactly, for Markov-chain data, in Chapters **??**–**??**. A heuristic that makes (4.8) plausible follows from the Fokker–Planck equation: rewriting its diffusion term as a transport term,

$$\frac{g^2}{2}\frac{\partial^2 p_t}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{g^2}{2}\, p_t\, \frac{\partial \log p_t}{\partial x}\right),$$

one sees that the density evolution of (4.1) is *identical* to that of the noise-free ODE with velocity $f - \frac{g^2}{2}\nabla \log p_t$ (the *probability-flow ODE* of Song et al., 2021); running that transport backwards, and re-injecting noise consistently, yields (4.8). The rigorous statement and proof are in Anderson (1982).

The chapter's summary is one sentence: *the OU channel* (4.4) *destroys data at a known rate, the Fokker–Planck equation describes the destruction at the level of densities, and Anderson's theorem says the destruction is invertible by anyone who knows the score* $\nabla \log p_t$ — which is why the rest of this thesis is about computing that score, exactly, when the data have Markov structure.

Chapter 5

Energy-Based Models, Markov Random Fields, and Factor Graphs

Chapter 2 derived the Boltzmann distribution from physics; this chapter adopts it as a *modelling language* for data. The three formalisms introduced here — energy-based models, Markov random fields, and factor graphs — are three views of the same object, ordered by increasing structural explicitness. The chapter is deliberately compact: every definition it contains is used, verbatim, in the research chapters, where the noisy Markov chain will be an energy-based model whose factor graph is a tree.

5.1 Energy-based models

An *energy-based model* (EBM) over $x \in \mathcal{X}$ is a probability distribution specified by an energy function E_θ:

$$p_\theta(x) \;=\; \frac{e^{-E_\theta(x)}}{Z(\theta)}, \qquad Z(\theta) = \int_{\mathcal{X}} e^{-E_\theta(x)}\,\mathrm{d}x, \tag{5.1}$$

the Boltzmann distribution (2.9) with β absorbed into the energy and the physics replaced by a parametric family (Ackley et al., 1985; LeCun et al., 2006). The appeal of (5.1) is that E_θ is unconstrained — any function defines a valid unnormalised density; the price is the partition function. Every basic operation hits it:

- *Evaluation* of $p_\theta(x)$ requires $Z(\theta)$ — intractable in general (Section 2.5).
- *Sampling* requires MCMC (Chapter 3), whose mixing time can be exponential in the presence of the metastability discussed in Section 2.5.
- *Learning* by maximum likelihood requires the gradient

$$-\nabla_\theta \log p_\theta(x) = \nabla_\theta E_\theta(x) - \mathbb{E}_{p_\theta}\Big[\nabla_\theta E_\theta\Big], \tag{5.2}$$

  whose second term — the model expectation — again demands sampling from p_θ.

There are two classical escape routes from the partition function, and this thesis walks both. The first is the *score*: since $\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$, the score is free of Z — differentiating in x kills the constant. Score matching (Chapter 6) and diffusion models are built on this observation. The second is *structure*: when the energy decomposes into local terms, marginals and even Z itself can become tractable by dynamic programming — that route is the rest of this chapter.

5.2 Markov random fields

Definition 5.1 (Markov random field)**.** Let $G = (V, \mathcal{E})$ be an undirected graph with one random variable x_i per node. The collection $x = (x_i)_{i \in V}$ is a *Markov random field* (MRF) if it satisfies the local Markov property: for every node i,

$$p\Big(x_i \,\Big|\, x_{V\setminus\{i\}}\Big) \;=\; p\Big(x_i \,\Big|\, x_{\partial i}\Big), \tag{5.3}$$

where ∂i is the set of neighbours of i in G.

The fundamental structure theorem connects this conditional-independence definition to the energy picture: by the Hammersley–Clifford theorem (proved in Besag, 1974), a strictly positive density is an MRF on G if and only if it factorises over the *cliques* of G,

$$p(x) \;=\; \frac{1}{Z} \prod_{C \in \mathrm{cliques}(G)} \psi_C(x_C) \;=\; \frac{1}{Z} \exp\Big[-\sum_C E_C(x_C)\Big]. \tag{5.4}$$

Conditional independence, graphical locality, and additive local energies are one and the same thing. The Ising model (2.12) is the textbook example: cliques are edges, and $E_{(i,j)} = -J_{ij}\sigma_i\sigma_j$.

For *our* purposes the decisive example is one-dimensional. The Markov chain factorisation (3.2), $p(x) = p_0(x_0)\prod_k M(x_{k+1}|x_k)$, is of the form (5.4) with G a path graph: a Markov chain is an MRF on a line, with energy

$$E(x) \;=\; -\log p_0(x_0) \;-\; \sum_{k=0}^{K-2} \log M(x_{k+1} \,|\, x_k), \tag{5.5}$$

a sum of *pairwise, nearest-neighbour* terms. For the Gaussian AR(1) chain each term is quadratic, so E is a quadratic form whose coupling matrix connects only neighbours — read: a *tridiagonal precision matrix.* Chapter **??** derives this precision explicitly; the reader should note that its tridiagonality is not an accident of Gaussian algebra but the Hammersley–Clifford fingerprint of one-dimensional Markov structure.

5.3 Hidden Markov models: chains observed through noise

The research problem of this thesis is, structurally, a *hidden Markov model* (HMM) (Rabiner, 1989): a latent Markov chain $(a_0, \ldots, a_{K-1})$ evolving by (3.6), observed only through independent noisy measurements — in our case, one OU channel (4.4) per frame,

$$x_k \;=\; a_k\, e^{-t} + \sqrt{\Delta_t}\, \varepsilon_k, \qquad \varepsilon_k \text{ i.i.d. } \mathcal{N}(0,1). \tag{5.6}$$

The joint law of latents and observations inherits both factorisations:

$$p(a, x) \;=\; \underbrace{p_0(a_0) \prod_{k=0}^{K-2} M(a_{k+1}|a_k)}_{\text{chain prior}} \cdot \underbrace{\prod_{k=0}^{K-1} \mathcal{N}\Big(x_k;\, a_k e^{-t}, \Delta_t\Big)}_{\text{local likelihoods}}. \tag{5.7}$$

Conditioning on the observations gives the *posterior* $p(a|x) \propto p(a, x)$, which is again an MRF on the line: the likelihood factors touch one variable each and therefore do not enlarge any clique. This innocent remark is the structural core of the thesis, for two reasons. First, as shown in Chapter 6, the diffusion score of the noisy sequence is exactly a posterior expectation under (5.7) — so computing scores *is* HMM inference. Second, posterior inference on chain-shaped MRFs is solvable in $O(K)$ by dynamic programming — the forward–backward algorithm for discrete HMMs (Rabiner, 1989), the Kalman smoother for linear-Gaussian ones (Kalman, 1960; Rauch et al., 1965) — and both are special cases of the single algorithm this thesis studies: belief propagation.

5.4 Factor graphs

MRFs record *which* variables interact but not *how* the joint factorises (a triangle of pairwise terms and one triple term have the same graph). The representation that makes factorisation itself graphical — and on which message passing is most naturally defined — is the *factor graph* (Kschischang et al., 2001).

Definition 5.2 (Factor graph)**.** Given a factorisation $p(x) = \frac{1}{Z}\prod_a \psi_a(x_{\partial a})$, its factor graph is the bipartite graph with a *variable node* for each x_i, a *factor node* for each ψ_a, and an edge (i, a) whenever $x_i \in x_{\partial a}$ (i.e. variable i appears in factor a).

A factor graph is a *tree* when it is connected and cycle-free; trees are the domain where the sum–product algorithm of Chapter **??** computes every marginal exactly (Pearl, 1988; Mézard and Montanari, 2009; Bishop, 2006).

The posterior (5.7) of our noisy chain has factor graph: variable nodes $a_0, \ldots, a_{K-1}$; pairwise factor nodes $M_k = M(a_{k+1}|a_k)$ between consecutive variables; and a unary factor node $L_k = \mathcal{N}(x_k; a_k e^{-t}, \Delta_t)$ hanging off each variable (the observations x_k are fixed numbers, not nodes). Pictorially it is a *caterpillar*: a spine of K variables alternating with $K-1$ transition factors, plus one likelihood leaf per vertebra. Removing any edge disconnects it; it contains no cycle; it is a tree. Hence — recording the conclusion the research chapters will exploit —

Proposition 5.3. *For any Markov-chain prior and any per-frame observation model, the posterior $p(a \mid x)$ factorises over a tree-shaped factor graph. Exact marginal inference on it costs $O(K)$ message-passing operations.*

Note what Proposition 5.3 does *not* say: it does not say the messages are finitely parametrisable. For discrete chains they are vectors; for the Gaussian chain they close within the Gaussian family (Chapter **??**); for general continuous chains they are infinite-dimensional functions, and *that* — not the graph topology — is where approximation enters. Keeping "tree-exactness of the algorithm" separate from "representability of the messages" is the conceptual hinge of this thesis, and the distinction the Gaussian closure results of Chapter **??** are designed to quantify.

5.5 What learning adds, and why we go exact

A learned generative model of sequences — an EBM with a neural energy, or the diffusion models of the next chapter — must discover the structure that equations (5.5)–(5.7) hand us for free. The modelling bet of this thesis is that the exactly solvable case is informative about the learned one: if even the *exact* posterior develops long-range couplings as noise grows (Chapter **??** makes this precise), then a learned score faces the same fate; and if exact inference degrades gracefully under locality constraints (Chapter **??**), then local architectures are a principled bet, not just a cheap one. The next chapter completes the toolkit by explaining how modern diffusion models train against these objects at scale.

Chapter 6

Diffusion Models: from Thermodynamics to Scores

This chapter assembles the material of Chapters 2–5 into the modern theory of generative diffusion. We present the discrete-time construction (DDPM), the score-matching principle that makes training tractable, the continuous-time SDE formulation that unifies both, and the identity — Tweedie's — that converts the score into a Bayesian posterior expectation. The chapter closes with the statistical-physics analyses of diffusion that motivate this thesis, and states the gap our research chapters fill.

6.1 The generative problem and the diffusion idea

Given samples of an unknown distribution p_0 on $\mathbb{R}^K$, a generative model must produce new samples. Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021) solve this with a two-process design inspired by non-equilibrium thermodynamics: a fixed *forward process* gradually corrupts data into pure noise, and a learned *reverse process* undoes the corruption step by step. The forward process used throughout this thesis is the OU channel of Section 4.2: at diffusion time t,

$$x_t \;=\; a\,e^{-t} + \sqrt{\Delta_t}\,\varepsilon, \qquad a \sim p_0, \quad \varepsilon \sim \mathcal{N}(0, I), \qquad \Delta_t = 1 - e^{-2t}, \tag{6.1}$$

so that the noised marginal is the convolution

$$p_t(x) \;=\; \int p_0(a)\, \mathcal{N}\Big(x;\, ae^{-t},\, \Delta_t I\Big)\, \mathrm{d}a. \tag{6.2}$$

As t grows, p_t interpolates from the data distribution to $\mathcal{N}(0, I)$. By Anderson's theorem (Section 4.4), the corruption is undone by the reverse SDE (4.8), whose only unknown ingredient is the score $\nabla_x \log p_t(x)$ of the noised marginals. Generative diffusion is therefore, at bottom, *score estimation*: whoever knows $\nabla \log p_t$ for all t can sample from p_0.

6.2 Denoising diffusion probabilistic models

The discrete-time formulation (Sohl-Dickstein et al., 2015; Ho et al., 2020) discretises (6.1) into a Markov chain $q(x_s|x_{s-1}) = \mathcal{N}(x_s; \sqrt{1-\beta_s}\, x_{s-1}, \beta_s I)$ with a variance schedule $\beta_1, \ldots, \beta_S$, and learns a reverse chain $p_\theta(x_{s-1}|x_s)$. Training maximises a variational lower bound on the data log-likelihood — the physicist will recognise the Gibbs variational principle (2.11) in modern dress.

Toolbox 6.0 — The evidence lower bound of a diffusion model

For any latent-variable model with joint $p_\theta(x_{0:S})$ and any inference distribution $q(x_{1:S}|x_0)$, Jensen's inequality gives

$$\log p_\theta(x_0) = \log \int p_\theta(x_{0:S})\, \mathrm{d}x_{1:S} = \log \mathbb{E}_q\left[\frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)}\right] \;\geq\; \mathbb{E}_q\left[\log \frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)}\right] =: \mathrm{ELBO}.$$

Insert the two Markov factorisations $p_\theta(x_{0:S}) = p(x_S) \prod_s p_\theta(x_{s-1}|x_s)$ and $q(x_{1:S}|x_0) = \prod_s q(x_s|x_{s-1})$, then regroup telescopically using Bayes' rule $q(x_s|x_{s-1}) = q(x_{s-1}|x_s, x_0)\, q(x_s|x_0)/q(x_{s-1}|x_0)$:

$$-\mathrm{ELBO} = \underbrace{D_{\mathrm{KL}}(q(x_S|x_0)\,\|\,p(x_S))}_{\text{prior term}} + \sum_{s=2}^{S} \underbrace{\mathbb{E}_q\, D_{\mathrm{KL}}(q(x_{s-1}|x_s, x_0)\,\|\,p_\theta(x_{s-1}|x_s))}_{\text{one denoising step each}} - \mathbb{E}_q \log p_\theta(x_0|x_1).$$

The decisive point: because the forward chain is Gaussian and conditioned on *both* endpoints, each target $q(x_{s-1}|x_s, x_0)$ is an explicit Gaussian. Matching it with a Gaussian p_θ reduces each KL term to a weighted regression of the posterior mean — concretely, to predicting the noise ε that was added (Ho et al., 2020). The thermodynamic reading of the ELBO gap as an entropy-production bound goes back to Sohl-Dickstein et al. (2015). $\square$

The practical upshot of Ho et al. (2020) is the loss

$$\mathcal{L}(\theta) = \mathbb{E}_{s,\, a\sim p_0,\, \varepsilon} \Big\| \varepsilon - \varepsilon_\theta\Big(\underbrace{a\, e^{-t_s} + \sqrt{\Delta_{t_s}}\, \varepsilon}_{x_{t_s}},\ s\Big)\Big\|^2, \tag{6.3}$$

a denoising regression. Its connection with the score is not incidental, as the next section shows: predicting the noise *is* estimating $\nabla \log p_t$ up to scale.

6.3 Score matching

Suppose we model the score directly by a function $s_\theta(x)$ and wish it to match $\nabla \log p(x)$ for data from p. The naive objective $\mathbb{E}_p \|s_\theta - \nabla \log p\|^2$ seems to require knowing the very score we lack. Two classical results remove the obstruction.

Toolbox 6.1 — Implicit score matching (Hyvärinen)

Expand the square and keep only θ-dependent terms:

$$\mathbb{E}_p \left\|s_\theta - \nabla \log p\right\|^2 = \mathbb{E}_p \left\|s_\theta\right\|^2 - 2\, \mathbb{E}_p \left\langle s_\theta, \nabla \log p \right\rangle + \text{const.}$$

The cross term integrates by parts. Componentwise, using $p\, \partial_i \log p = \partial_i p$ and assuming $p\, s_{\theta,i} \to 0$ at infinity:

$$\int p\ s_{\theta,i}\ \partial_i \log p \ \mathrm{d}x = \int s_{\theta,i}\ \partial_i p\ \mathrm{d}x = -\int p\ \partial_i s_{\theta,i}\ \mathrm{d}x.$$

Therefore

$$\mathbb{E}_p \|s_\theta - \nabla \log p\|^2 = \mathbb{E}_p\Big[\|s_\theta(x)\|^2 + 2\,\nabla \cdot s_\theta(x)\Big] + \text{const} :$$

an objective involving only the model and data samples (Hyvärinen, 2005). Elegant, but the divergence term is expensive for large networks — which is why diffusion models use the denoising variant instead. □

Toolbox 6.2 — Denoising score matching (Vincent)

Fix t and consider the noised marginal (6.2), written as $p_t(x) = \int p_0(a)\, q_t(x|a)\,\mathrm{d}a$ with Gaussian kernel $q_t(x|a) = \mathcal{N}(x; ae^{-t}, \Delta_t I)$. Then

$$\nabla_x \log p_t(x) = \frac{\int p_0(a)\,\nabla_x q_t(x|a)\,\mathrm{d}a}{p_t(x)} = \int \underbrace{\frac{p_0(a)\,q_t(x|a)}{p_t(x)}}_{=p(a|x)} \nabla_x \log q_t(x|a)\;\mathrm{d}a = \mathbb{E}\Big[\nabla_x \log q_t(x|a)\;\Big|\;x\Big].$$

The score of the *marginal* is the posterior average of the score of the *kernel* — and the latter is trivial: $\nabla_x \log q_t(x|a) = -(x - ae^{-t})/\Delta_t$. A standard L^2-projection argument turns this into a regression: for any s_θ,

$$\mathbb{E}_{x\sim p_t} \|s_\theta(x) - \nabla \log p_t(x)\|^2 = \mathbb{E}_{a,x} \|s_\theta(x) - \nabla_x \log q_t(x|a)\|^2 + \text{const},$$

because conditional expectation is exactly the L^2 projection onto functions of x (Vincent, 2011). Substituting the Gaussian kernel score and $x = ae^{-t} + \sqrt{\Delta_t}\varepsilon$ gives $\nabla_x \log q_t = -\varepsilon/\sqrt{\Delta_t}$: *learning the score of noisy data is learning to predict the noise*, and the DDPM loss (6.3) is denoising score matching in disguise. □

The continuous-time synthesis (Song and Ermon, 2019; Song et al., 2021) trains one network $s_\theta(x,t)$ against denoising score matching at all noise levels and samples with the reverse SDE (4.8) or its deterministic twin, the probability-flow ODE; an accessible exposition is Song (2021), and Karras et al. (2022) systematise the design space.

6.4 Tweedie's identity: the score is a denoiser

The middle line of Vincent's derivation deserves its own section, because it is the bridge on which the whole thesis walks. Rearranging $\nabla \log p_t(x) = \mathbb{E}[-(x - ae^{-t})/\Delta_t \mid x]$:

$$\boxed{\nabla_x \log p_t(x) \;=\; \frac{e^{-t}\,\mathbb{E}[\,a\,|\,x\,] - x}{\Delta_t} \qquad \Longleftrightarrow \qquad \mathbb{E}[\,a\,|\,x\,] = e^{t}\Big(x + \Delta_t\,\nabla_x \log p_t(x)\Big).} \tag{6.4}$$

This is the Tweedie–Miyasawa formula of empirical Bayes (Robbins, 1956; Miyasawa, 1961; Efron, 2011): the score of the noisy marginal and the Bayes-optimal denoiser $\mathbb{E}[a|x]$ determine each other exactly, with no assumption whatsoever on p_0. Three readings will recur:

1. *Geometric*: the score points from x towards (the contracted image of) the posterior mean of the clean data — denoising direction and score direction coincide.

2. *Statistical*: estimating the score is solving a Bayesian inference problem — compute a posterior expectation under the joint (5.7). Everything Chapter 5 said about inference on graphs now applies to scores.

3. *Algorithmic*: for sequence data with Markov prior, the posterior lives on a tree (Proposition 5.3), so the score is computable by message passing — the programme of Chapter **??**.

For *vector-valued* data $a \in \mathbb{R}^K$ noised coordinate-wise, (6.4) holds componentwise with the *full* posterior:

$$\Big(\nabla_x \log p_t(x)\Big)_k \;=\; \frac{e^{-t}\,\mathbb{E}[\,a_k\,|\,x_0,\ldots,x_{K-1}\,] - x_k}{\Delta_t}. \tag{6.5}$$

The conditioning is on *all* coordinates: even though the noise is independent across frames, the posterior mean of frame k — and hence the k-th score component — depends on every observation, because the prior couples the frames. Equation (6.5) is the precise reason the *joint* score differs from the per-frame marginal score, a distinction that is easy to blur and whose importance for structured data was impressed on this project in supervision; Chapter **??** quantifies exactly how much coupling the conditioning induces, as a function of t.

6.5 The statistical physics of generative diffusion

A recent line of work analyses diffusion models with the tools of Chapter 2, in the regime where the score is *exact* — either computed from a solvable p_0 or from the empirical measure of n training points. Three findings frame our work.

Regimes and transitions. For high-dimensional solvable data (Gaussian mixtures, spin models), the reverse dynamics crosses sharp dynamical regimes (Biroli and Mézard, 2023; Biroli et al., 2024): a *speciation* time at which the trajectory commits to one mode — a spontaneous symmetry breaking in the sense of Section 2.5 (see also Raya and Ambrogioni, 2023) — and, when the score is learned from finitely many samples, a *memorisation* (collapse) time below which the dynamics condenses onto training points. The phase boundaries are computed with free-energy methods, in direct analogy with the random energy model.

Sampling as physics. Ghio et al. (2024) map generative samplers — flows, diffusions, autoregressive networks — onto statistical-physics problems and delimit where each is efficient, importing the algorithmic-hardness picture of spin glasses (Zdeborová and Krzakala, 2016).

The exactly solvable programme. The common method of these works is to choose p_0 rich enough to be interesting and simple enough that $\nabla \log p_t$ is available in closed form, then study the *exact* generation dynamics. What none of them vary is the *internal structure of a single data point*: their coordinates are exchangeable or i.i.d. conditional on a mode; there is no ordering, no dynamics *inside* the sample.

6.6 The gap, restated

Real sequential data — video, audio, trajectories — have exactly the structure the solvable programme leaves out: an internal (frame) time with a Markovian dependence along it. For such data the score (6.5) is a smoothing posterior on a chain, an object with seventy years of inference theory behind it (Chapters 3–5). Yet the questions a diffusion modeller asks about it — how does the coupling structure of $\nabla \log p_t$ deform with t? can it be computed *locally*? what survives a Gaussian parametrisation of the messages when the data are not Gaussian? what does that imply for score architectures? — are not answered by the classical literature, which fixes the noise level, nor by the diffusion literature, which ignores internal structure. Answering them, in the fully solvable setting of the AR(1) chain and its non-Gaussian perturbations, is the contribution of Chapters **??**–**??**.

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